

Einstein's General Theory of Relativity

Lecture 1: Newtonian Gravity, Free Fall, and Gauss's Theorem

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Chapter 1

Newtonian Gravity, Free Fall, and Gauss's Theorem

1.1 Why Begin General Relativity with Newton?

Gravity is unlike the electric and magnetic forces. Its deepest formulation is tied to the geometry of space and time, and that connection is the subject of general relativity. We shall not begin there. We shall begin with Newtonian gravity, but arrange the old theory so that the ideas needed later are already visible.

The natural starting point is an inertial frame: a set of spatial coordinates and clocks in which a force-free test body moves with constant velocity. In such a frame Newton's equation is

$$\vec{F} = m\vec{a}. \quad (1.1)$$

The force and acceleration are three-vectors; the mass is a scalar. In components,

$$F_i = ma_i = m\ddot{x}_i, \quad i = 1, 2, 3. \quad (1.2)$$

A particle's coordinates x_1, x_2, x_3 are the components of its position vector $\vec{r}$. Their first and second time derivatives are the components of velocity and acceleration:

$$\vec{v} = \dot{\vec{r}}, \quad \vec{a} = \ddot{\vec{r}}. \quad (1.3)$$

In Newtonian mechanics mass is conserved. A cup becomes lighter when coffee is removed, but that is transport of matter, not spontaneous loss of mass. These elementary distinctions will matter because gravity places the same mass on both sides of Equation (1.1).

1.2 Galileo's Uniform Gravitational Field

Before using Newton's inverse-square law, take the approximation appropriate to experiments near the Earth's surface. Over a sufficiently small region the Earth may be treated as flat. Then *down* has the same direction everywhere, and the strength of gravity may be treated as independent of height. If x_2 points upward, the vertical force is

$$F_2 = -mg. \quad (1.4)$$

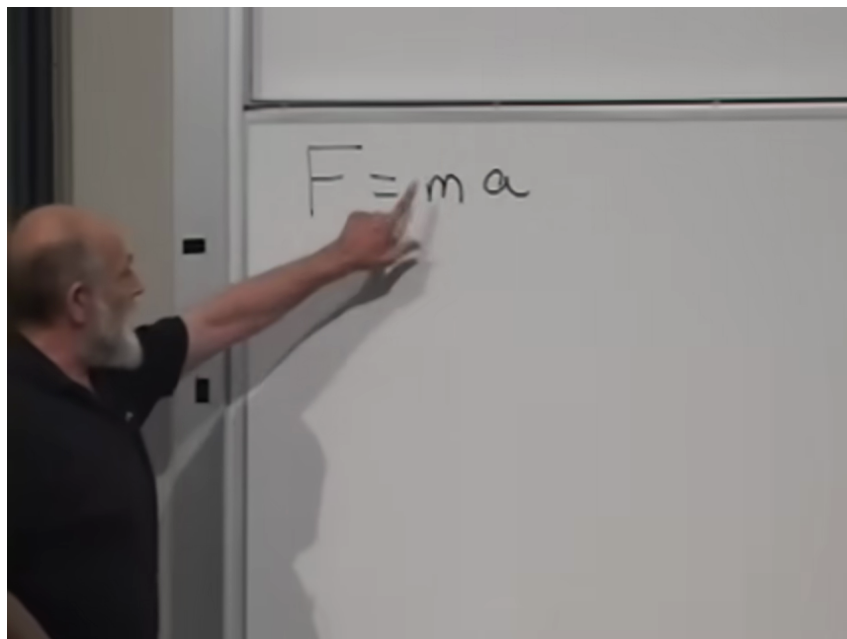


Figure 1.1: Newton's law is introduced before gravity is specialized. The spoken discussion immediately interprets the board equation as a three-vector relation.

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Here g is approximately 9.8 m s^{-2} at the Earth's surface. It differs on the Moon or Jupiter because it depends on the source, but it does not depend on the mass of the body being dropped.

Combining Equations (1.2) and (1.4),

$$m\ddot{x}_2 = -mg, \quad (1.5)$$

$$\boxed{\ddot{x}_2 = -g.} \quad (1.6)$$

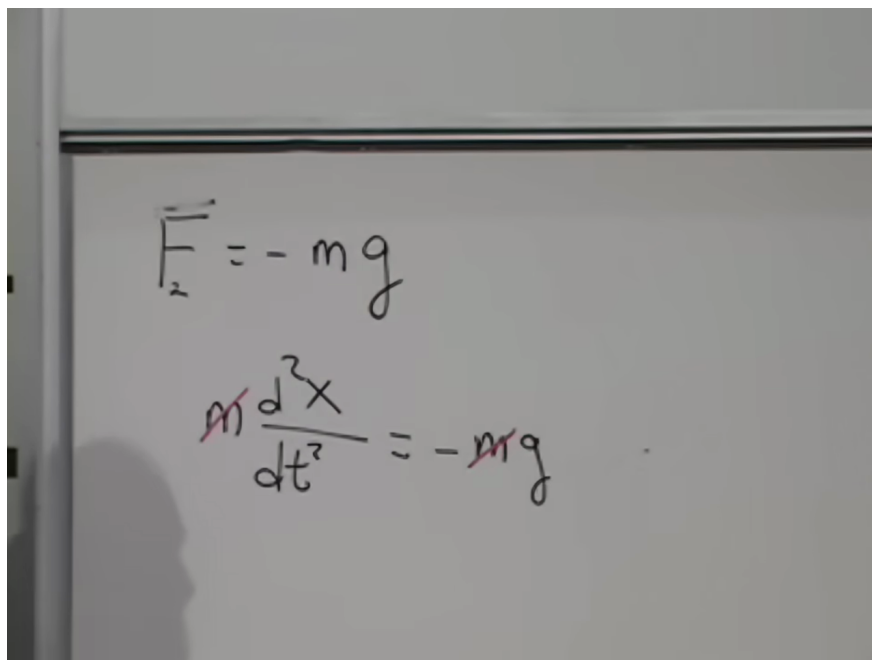
This cancellation is gravity's special feature. An electric force is proportional to charge, while inertia is proportional to mass; charged and uncharged objects therefore respond differently to an electric field. In gravity the coupling and the inertia are proportional to the same quantity. Absent air resistance, a cannonball and a feather obey the same trajectory. Whether Galileo actually dropped objects from the Leaning Tower of Pisa is less important than the content of the experiment; inclined-plane measurements already expose the same universal acceleration.

1.3 Free Fall and the First Equivalence Principle

The cancellation has a stronger consequence than equal fall times. Imagine a diffuse cloud of noninteracting particles released from rest. The particles may have different masses and sizes and may begin at different heights. In a uniform field each obeys $\ddot{\vec{r}}_a = \vec{g}_0$, so for any pair,

$$\frac{d^2}{dt^2} (\vec{r}_a - \vec{r}_b) = 0. \quad (1.7)$$

If their initial relative velocities vanish, their separations remain fixed. The cloud translates without changing shape.



A whiteboard with two equations written on it. The first equation is $\vec{F}_2 = -mg$. The second equation is $m \frac{d^2x}{dt^2} = -mg$, where the m on the left and the m on the right are crossed out with red ink.

Figure 1.2: Galileo's near-surface law and the cancellation of the falling body's mass from its equation of motion.

Lecture frame: 00:12:40.



Figure 1.3: A loose cloud of particles used to make universality of free fall into a statement about relative motion and internal stress.

Lecture frame: 00:17:00.

Even if struts held the particles together, the struts would carry no gravity-induced stress in this approximation: every part wants to make the same motion. A self-contained freely falling laboratory is therefore mechanically indistinguishable from an inertial laboratory far from all gravitating bodies. This is the first form of the equivalence principle.

Question from the audience. Could the falling cloud detect gravity optically, perhaps by measuring laser distances between its particles? ^a

Response. Not in a perfectly uniform field. The relative positions of freely falling particles do not change, so internal optical measurements give the same result as they would in gravity-free inertial motion. Looking at the floor would reveal motion relative to an external supported object, but that is no longer a self-contained experiment. The floor fails to fall only because other forces support it.

^aLecture timestamp: 00:17:30–00:20:16.

Question from the audience. Surely we can feel when we are accelerating. Does the peculiar sensation of stepping from a high building reveal gravity? ^a

Response. What the body ordinarily feels is the support force that prevents free fall: pressure on the feet, the chair, and the internal forces transmitting that pressure. Remove the support and those forces disappear. The resulting sensation would be the same in a freely drifting spacecraft far from gravitating bodies. In this precise sense we feel *not falling*, rather than uniform free fall itself.

^aLecture timestamp: 00:20:17–00:22:44.

The qualification *uniform* is essential. A real planet is finite and round. Its field changes with position, and those changes will soon restore an observable effect through tidal forces.

1.4 Newton's Inverse-Square Law

Move now beyond the flat-Earth approximation. Every mass attracts every other mass. For two bodies of masses m and M , separated by a distance r , the magnitude of either force is

$$F = \frac{GmM}{r^2}, \quad (1.8)$$

where

$$G \simeq 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}. \quad (1.9)$$

The constant is needed dimensionally, and its small numerical value records the weakness of gravity on laboratory scales. Two one-kilogram masses one meter apart attract with only about 6.67×10^{-11} N. A rubbed, electrically charged object or a small magnet produces a readily visible deflection where even a nearby heavy mass does not. Gravity dominates our weight because the Earth supplies an enormous source mass.

Question from the audience. If the two forces are equal and opposite, why does a falling object accelerate noticeably while the Earth hardly moves? ^a

Response. Newton's third law equates the forces, not the accelerations. Since $\vec{a} = \vec{F}/m$, the same force gives the enormous Earth an imperceptibly small acceleration and gives the much lighter object a much larger one. Both bodies actually move about their common center of mass.

^aLecture timestamp: 00:27:07–00:29:02.

1.4.1 Kepler's Clue

Newton did not obtain the inverse-square dependence from dimensional analysis. Kepler's planetary laws supplied the decisive clue. For a circular orbit of radius r and angular frequency ω ,

$$a_{\text{circ}} = \omega^2 r. \quad (1.10)$$

If the acceleration caused by gravity varies as $1/r^2$, then

$$\omega^2 r \propto \frac{1}{r^2} \implies \omega^2 \propto \frac{1}{r^3}. \quad (1.11)$$

Since $T = 2\pi/\omega$,

$$\boxed{T^2 \propto r^3}, \quad (1.12)$$

which is Kepler's period-radius law. The same inverse-square force also produces elliptical, not merely circular, bound orbits.

Question from the audience. What happens as the two objects approach one another or touch? Does the inverse-square law diverge? ^a

Response. The Newtonian point-particle expression is not a complete short-distance theory of matter. Once extended bodies touch, electrostatic, atomic, and nuclear forces become important. The contact force supporting an eraser on a table is itself largely electromagnetic. Newton's law remains the long-range gravitational contribution, not the only interaction.

^aLecture timestamp: 00:32:32–00:34:37.

This comparison also sharpens the universality of gravity. Electric fields distinguish charge from neutrality and can attract or repel. Newtonian gravity acts on every mass and is always attractive.

1.5 Direction and the Many-Body Law

Equation (1.8) gives only a magnitude. Put M at the origin and let $\vec{r}$ point from M to m . Attraction makes the force point opposite to $\vec{r}$:

$$\vec{F} = -\frac{GmM}{r^2}\hat{r} = -\frac{GmM}{r^3}\vec{r}, \quad \hat{r} = \frac{\vec{r}}{r}. \quad (1.13)$$

The extra r in the denominator compensates for the magnitude of the vector inserted in the numerator.

For many particles, let $\vec{r}_i$ and m_i be the position and mass of particle i . Define $\vec{r}_{ij} = \vec{r}_j - \vec{r}_i$, the vector pointing from i to j . The force on i is then

$$\vec{F}_i = \sum_{j \neq i} Gm_i m_j \frac{\vec{r}_j - \vec{r}_i}{|\vec{r}_j - \vec{r}_i|^3}. \quad (1.14)$$

Every term points toward its source. The lecture's temporary minus-sign confusion is only a choice between $\vec{r}_{ij}$ and $\vec{r}_{ji} = -\vec{r}_{ij}$; once the displacement vector is defined, the sign is fixed.

Substitution into $m_i \vec{a}_i = \vec{F}_i$ cancels the mass of the body whose motion is being followed:

$$\boxed{\vec{a}_i = \sum_{j \neq i} Gm_j \frac{\vec{r}_j - \vec{r}_i}{|\vec{r}_j - \vec{r}_i|^3}}. \quad (1.15)$$

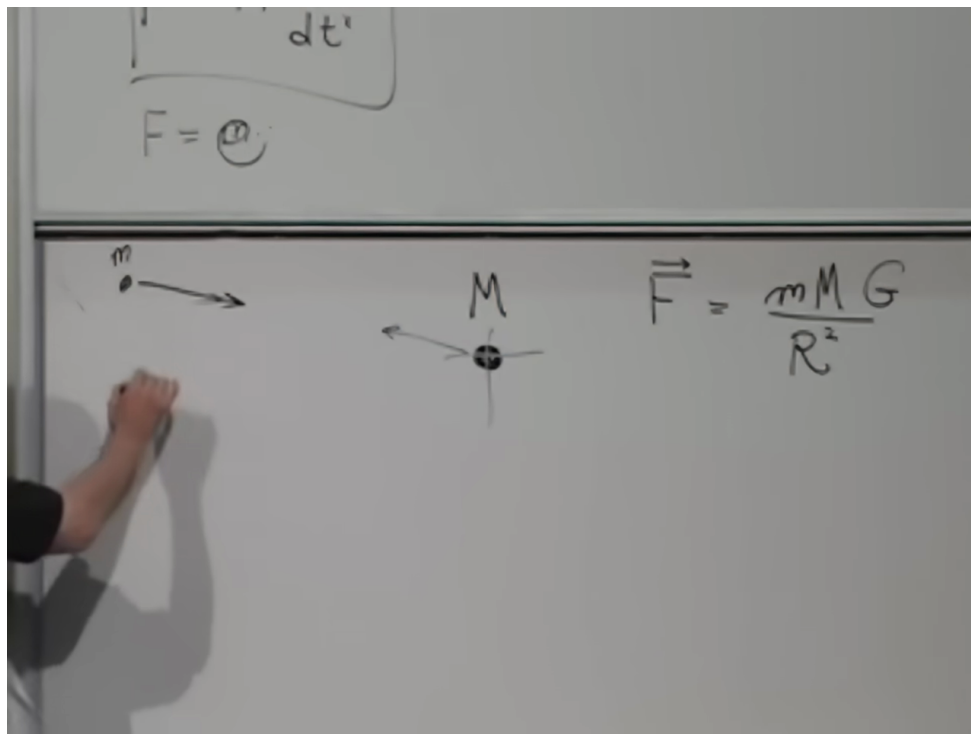


Figure 1.4: The scalar inverse-square law and the source-to-particle geometry from which its vector direction is fixed.

Lecture frame: 00:36:25.

The acceleration depends on all the source masses but not on m_i . Universality of free fall survives the move from uniform gravity to the full Newtonian field.

Question from the audience. Once all the masses move, does every particle retain one uniform acceleration, perhaps toward the center of the whole system? ^a

Response. No. Equation (1.15) gives the instantaneous acceleration from the instantaneous positions. As the positions change, so do the magnitudes and directions of all terms. Bodies may even be moving apart while their accelerations point toward one another. Solving the coupled equations is a nonlinear many-body problem; superposition of instantaneous forces does not make the trajectories linear or uniform.

^aLecture timestamp: 00:43:28–00:46:43.

1.6 Tidal Forces: Where Uniform Free Fall Fails

A realistic gravitational field varies across an extended object. Imagine a person thousands of miles tall falling feet first toward the Earth. The feet are closer to the source and accelerate more strongly than the head, so the body stretches radially. Turn the person sideways. The forces on the two ends point along slightly different radii toward the Earth's center, so the body is compressed transversely.

For two nearby points separated by $\delta\vec{x}$,

$$\delta g_i = g_i(\vec{x} + \delta\vec{x}) - g_i(\vec{x}) \simeq \frac{\partial g_i}{\partial x_j} \delta x_j. \quad (1.16)$$

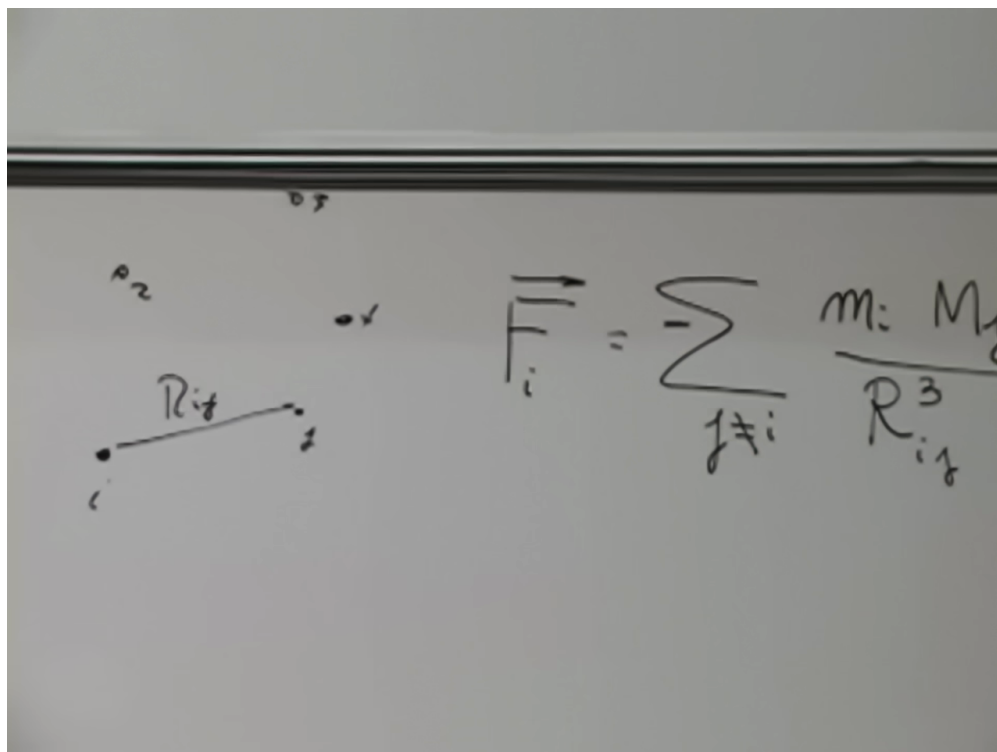


Figure 1.5: The many-body force sum together with the displacement-vector convention. This unobstructed frame precedes the live sign correction.

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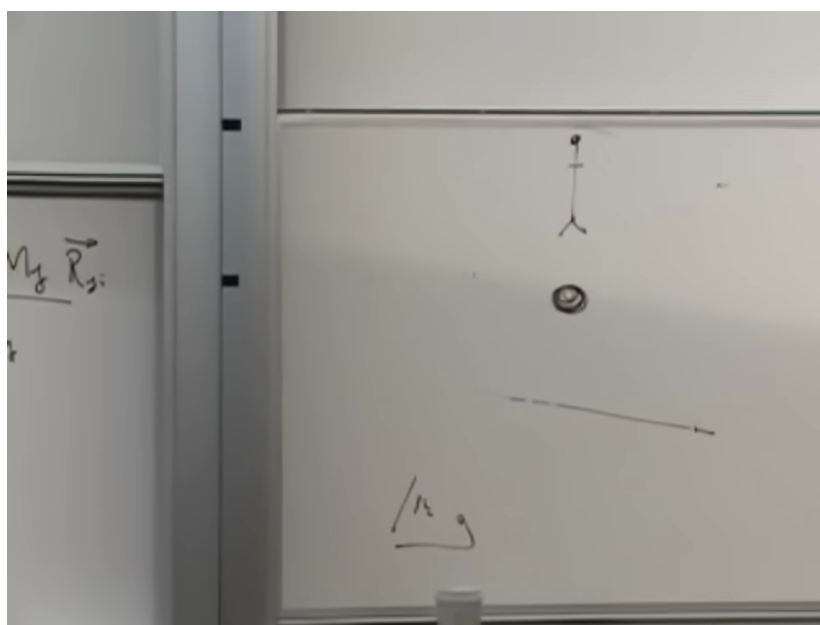


Figure 1.6: The board's extended-body sketches: radial variation stretches a vertical body, while converging field directions compress it transversely.

Lecture frame: 00:48:00.

Thus a freely falling laboratory removes the common acceleration $\vec{g}(\vec{x})$, but not its spatial gradient. For a point source,

$$\vec{g}(\vec{r}) = -\frac{GM}{r^3}\vec{r}. \quad (1.17)$$

The radial eigenvalue of its gradient is $2GM/r^3$, while each transverse eigenvalue is $-GM/r^3$: stretching in one direction is accompanied by compression in the other two.

These are tidal forces. The Moon's field is slightly stronger on the near side of the Earth and slightly weaker on the far side. The solid Earth resists deformation, while the oceans respond more visibly, producing approximately opposite tidal bulges. The effect becomes negligible only when the falling system is small compared with the length scale over which the field changes.

Question from the audience. Did Newton understand the lunar origin of the tides? ^a

Response. He understood the essential tidal mechanism, although the lecture does not settle how accurately he calculated the ocean's actual deformation. The point needed here is the differential pull: a finite body samples different values and directions of the Moon's field.

^aLecture timestamp: 00:50:29–00:51:09.

1.7 The Gravitational Field

The many-body formula suggests a cleaner description. Add an imagined test particle so light that it does not disturb the sources. Place it at $\vec{x}$ and observe its acceleration. The gravitational field is that acceleration:

$$\vec{g}(\vec{x}, t) \equiv \frac{\vec{F}_{\text{test}}}{m_{\text{test}}}. \quad (1.18)$$

Because the test mass cancels, the result is independent of which test particle is used. If source masses m_j occupy positions $\vec{r}_j(t)$,

$$\boxed{\vec{g}(\vec{x}, t) = \sum_j Gm_j \frac{\vec{r}_j(t) - \vec{x}}{|\vec{r}_j(t) - \vec{x}|^3}.} \quad (1.19)$$

It is a vector at every point in space and, when the sources move, a function of time as well.

Question from the audience. In a whole ensemble, do sign choices or mutual influence add some extra nonlinear term to the field formula? ^a

Response. At a specified instant the force is exactly the vector sum in Equation (1.14); attraction is already encoded by the direction $\vec{r}_j - \vec{x}$. Mutual influence enters when the positions are evolved, because each trajectory changes the future field seen by every other body. It complicates the solution, not the instantaneous force law.

^aLecture timestamp: 00:52:36–00:56:09.

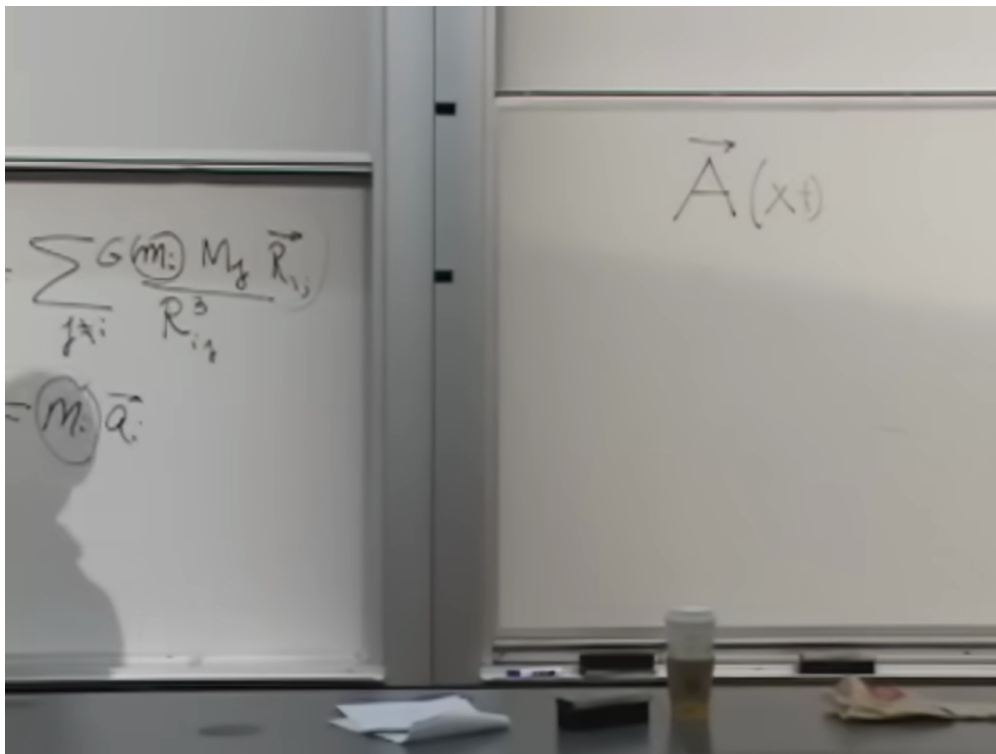


Figure 1.7: The test-particle acceleration emerges from the many-body force law after the test mass is canceled.

Lecture frame: 01:00:00.

The image shows a whiteboard with two handwritten equations. The top equation is $\vec{\nabla} \cdot \vec{A} = \partial_x A_x + \partial_y A_y + \partial_z A_z$. The bottom equation is $\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$.

Figure 1.8: The component definition of divergence. The arrows over ∇ and A matter: their dot product is a scalar.

Lecture frame: 01:08:10.

1.8 Divergence: Measuring Local Sources and Sinks

After the break we need one piece of vector calculus. For a generic vector field $\vec{A} = (A_x, A_y, A_z)$, define

$$\boxed{\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}.} \quad (1.20)$$

The upside-down triangle, ∇ , represents the three spatial derivatives. Although it acts on a vector field, the dot product leaves a scalar.

The name is easiest to understand through flow. Imagine a shallow lake of fixed depth filled with incompressible water. If water is pumped in from below, the horizontal velocity field spreads outward and has positive divergence. If water is removed through a hidden drain, the arrows converge and the divergence is negative. Uniform translation of the whole lake has zero divergence even though the water is moving.

The local test is to draw a small boundary. If more water leaves than enters, some water must have been supplied inside. In one dimension the same idea is visible when rightward arrows grow longer as x increases: $\partial A_x / \partial x > 0$.

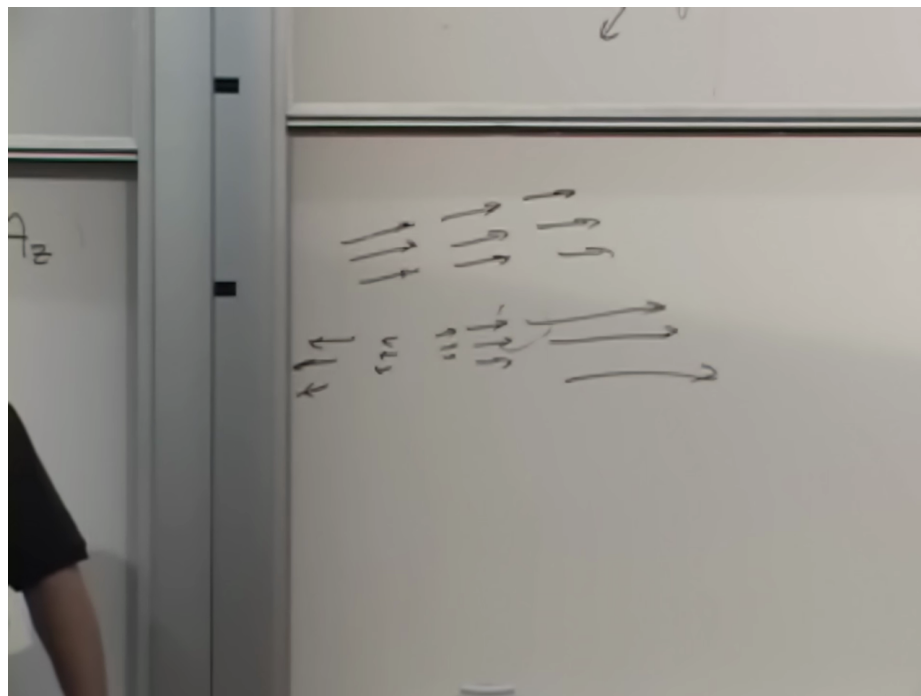


Figure 1.9: The arrow field used to compare inflow and outflow across a small imagined control region. Longer outgoing arrows indicate positive local divergence.

Lecture frame: 01:13:00.

Question from the audience. Could the arrows spread because a compressible fluid is thinning rather than because water is being pumped in? ^a

Response. Yes. For a compressible fluid, velocity divergence can change density instead of representing an external source. The lake analogy deliberately fixes both density and depth so that velocity divergence may be read directly as injected or removed water. The mathematical definition of divergence itself does not depend on that simplifying interpretation.

^aLecture timestamp: 01:12:18–01:14:45.

1.9 Gauss's Theorem

Let V be any volume with closed boundary ∂V and outward unit normal $\hat{n}$. Gauss's theorem states

$$\boxed{\int_V (\nabla \cdot \vec{A}) d^3x = \oint_{\partial V} \vec{A} \cdot \hat{n} d\sigma.} \quad (1.21)$$

The left side adds all local sources and sinks in the interior. The right side adds the net outward flux through the boundary. A tangential component does not cross the boundary and therefore contributes nothing.

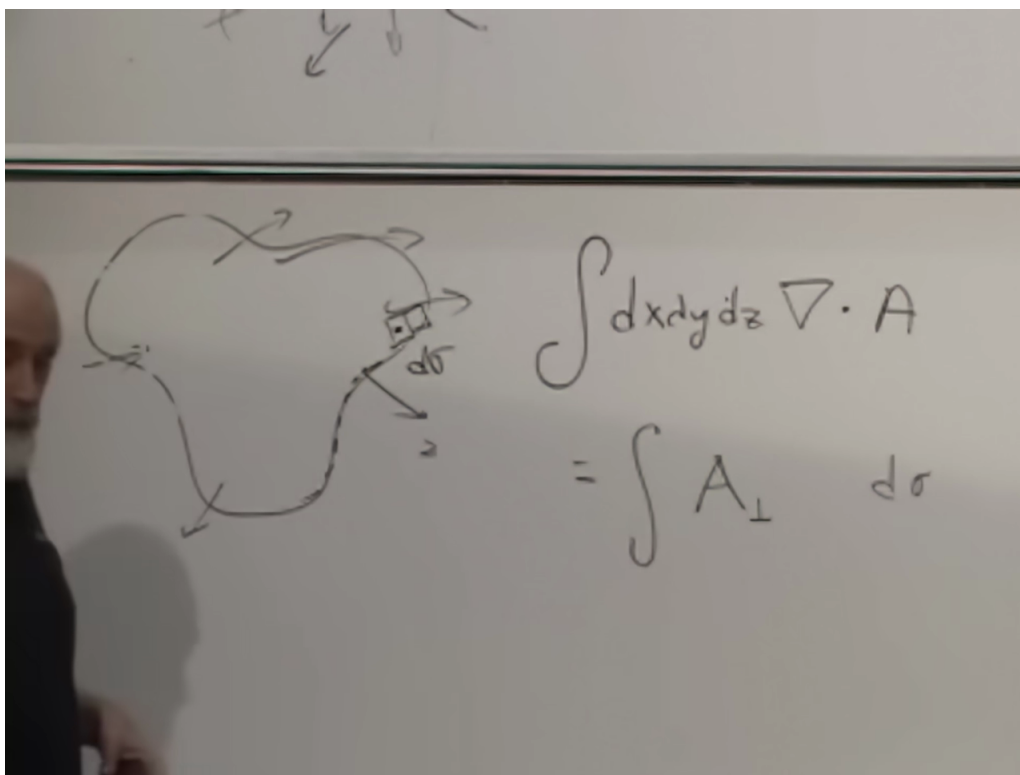


Figure 1.10: Gauss's theorem on the board: the volume integral of divergence equals the surface integral of the normal field component.

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Question from the audience. If no water is created, should the integrated divergence always vanish? ^a

Response. For an incompressible flow with no sources or sinks inside the chosen region, yes: its net outward flux is zero. Water may still enter one part of the boundary and leave another. A pump supplies positive divergence; a drain supplies negative divergence. In the two-dimensional lake the interior area integral equals a line integral around the boundary; in three dimensions it becomes Equation (1.21).

^aLecture timestamp: 01:20:37–01:22:33.

1.10 Spherical Symmetry and the Inverse-Square Field

Suppose $\nabla \cdot \vec{A}$ is confined to a finite spherically symmetric region. Surround it by a larger sphere of radius r , and define the total source strength

$$Q = \int_V (\nabla \cdot \vec{A}) d^3x. \quad (1.22)$$

As long as the Gaussian sphere encloses the source, Q is independent of its radius. Spherical symmetry requires the exterior field to be radial and to have the same magnitude at every point on the sphere. Therefore

$$Q = \oint_{\partial V} \vec{A} \cdot \hat{n} d\sigma \quad (1.23)$$

$$= A(r) 4\pi r^2, \quad (1.24)$$

and hence

$$\boxed{\vec{A}(r) = \frac{Q}{4\pi r^2} \hat{r}.} \quad (1.25)$$

The surface area is $4\pi r^2$; the transient phrase *four-thirds* in the spoken calculation is immediately corrected on the board.

Question from the audience. The result is written as a scalar magnitude. How is the vector direction restored? ^a

Response. Spherical symmetry supplies it. For positive divergence the field is radial and outward, so multiplying the magnitude by $\hat{r}$ gives Equation (1.25). For gravity the field points inward, and the sign is reversed.

^aLecture timestamp: 01:27:43–01:29:12.

Gravity is therefore analogous not to water emitted from a source but to incompressible flow drawn into a sink. In continuum notation the precise Newtonian statement is

$$\nabla \cdot \vec{g} = -4\pi G\rho, \quad (1.26)$$

so for enclosed mass $M_{\text{enc}} = \int_V \rho d^3x$,

$$\oint_{\partial V} \vec{g} \cdot \hat{n} d\sigma = -4\pi G M_{\text{enc}}. \quad (1.27)$$

For a point mass or any spherical source observed from outside,

$$\vec{g}(r) = -\frac{GM}{r^2} \hat{r}. \quad (1.28)$$

The fluid picture is only a mathematical analogy; it does not assert that space is literally flowing.

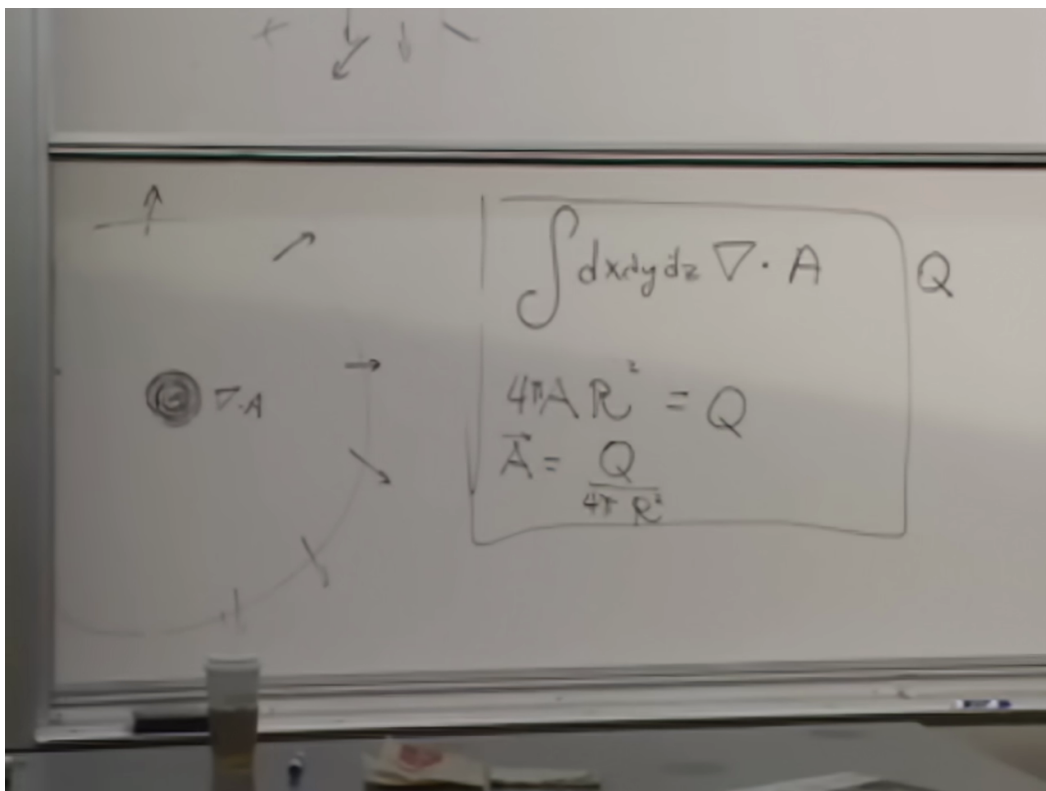


Figure 1.11: Spherical symmetry turns Gauss's theorem into $4\pi r^2 A(r) = Q$, forcing an inverse-square radial field.

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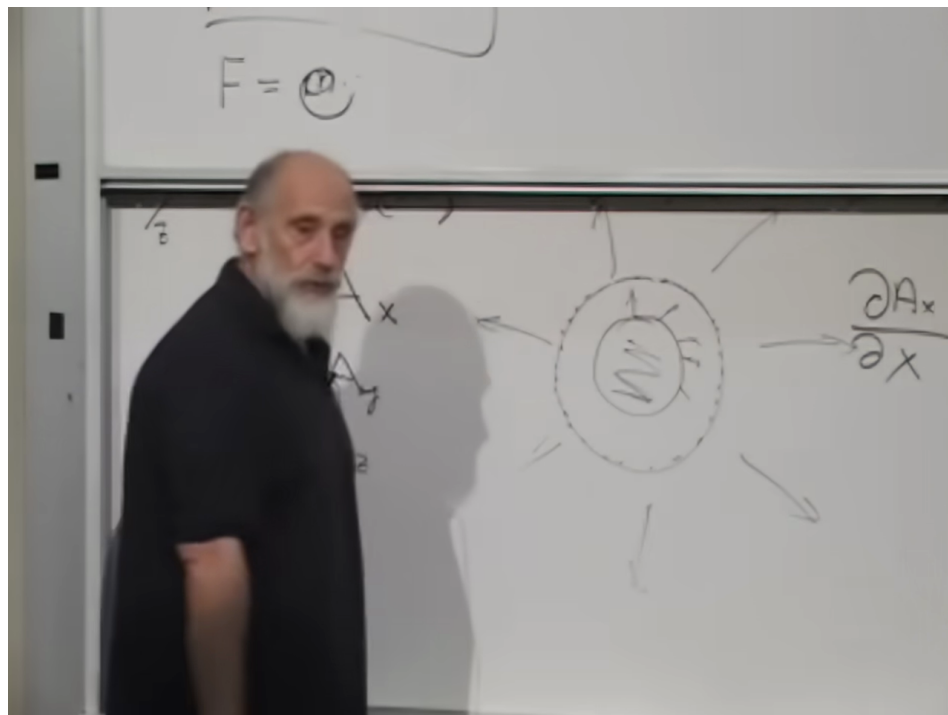


Figure 1.12: The hollow-shell discussion. Outside, the shell acts like a point mass at the center; inside, spherical symmetry and zero enclosed mass give zero field.

Lecture frame: 01:37:10.

1.11 Newton's Shell Theorem

The derivation used only spherical symmetry and total enclosed source. It did not use the detailed radial density profile. We have therefore proved Newton's theorem:

Theorem 1.1 (Newton's shell theorem). *Outside any spherically symmetric mass distribution, the gravitational field is exactly the field of a point mass with the same total mass placed at the center. Inside a spherical shell of matter, the gravitational field vanishes.*

The exterior statement makes planetary gravity tractable. A spherical Earth may be treated as though all its mass were at the center, provided the field point lies outside the material. This special simplification belongs to the inverse-square law; a generic $1/r^n$ force would not reproduce it.

The restriction to exterior points matters. In a mine shaft through a uniform-density Earth, a Gaussian sphere of radius $r < R$ encloses only

$$M_{\text{enc}}(r) = \frac{4\pi}{3} \rho r^3. \quad (1.29)$$

Consequently

$$\vec{g}(r) = -\frac{GM_{\text{enc}}(r)}{r^2} \hat{r} = -\frac{4\pi G\rho}{3} \vec{r}. \quad (1.30)$$

The field becomes small near the center and vanishes at the center itself; it is not the divergent field obtained by concentrating the entire Earth there.

Question from the audience. Inside a hollow shell, should the nearer wall not pull more strongly than the farther wall? ^a

Response. Individual nearby elements do pull more strongly, but there is less nearby area subtending a given solid angle; the larger amount of farther material cancels that advantage exactly. Gauss's theorem packages the cancellation. A centered Gaussian sphere encloses no mass, and spherical symmetry makes the radial field constant over it, so $4\pi r^2 g(r) = 0$ and therefore $\vec{g} = 0$ everywhere in the cavity.

^aLecture timestamp: 01:35:30–01:38:04.

General relativity has a corresponding exterior result: a spherically symmetric vacuum field is fixed by the total mass and has the Schwarzschild form, the same exterior geometry associated with a nonrotating black hole. That theorem lies ahead. For now the essential lesson is already visible in Newtonian language: universal free fall shifts attention from force to acceleration, spatial variation appears as tides, and Gauss's theorem turns local source information into the global field.